Li-700 (Adjuvant)

LI700 is a non-ionic surfactant used to improve spray coverage and penetration. Add to spray tanks as directed on the label. Mix thoroughly to ensure even distribution in the solution. LI700 enhances the effectiveness of herbicides, insecticides, and fertilizers by improving leaf coverage.

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023

1 Identification

Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

OtherMeansof Identification: Mixture

APVMA Approval Number: 66985

RecommendedUseoftheChemicalandRestriction on Use:
Adjuvant - acidifying and penetrating surfactant

Detailsof ManufacturerorImporter:
Nutrien Ag Solutions
Level 5, Building A
26 Talavera Road
Macquarie Park NSW 2113

Phone Number:(02) 9889 5400

Emergency telephone number:1800 033 111

2 Hazard(s) Identification

HazardousNature:
Classified as Hazardous according to the Globally Harmonised System of Classification and Labelling of Chemicals (GHS) and Safe Work Australia criteria.
Not classified as Dangerous Goods according to the Australian Code for the Transport of Dangerous Goods by Road and Rail (7th edition), IATA and IMDG/IMSBC.



Acute Toxicity (Oral) 4	H302Harmful if swallowed.
Eye Irritation 2A	H319Causes serious eye irritation.
STOT SE 3	H335May cause respiratory irritation.

Signal WordWarning

Hazard Statements
H302Harmful if swallowed.
H319Causes serious eye irritation.
H335May cause respiratory irritation.

Precautionary Statements

P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P264	Wash thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P271	Use only outdoors or in a well-ventilated area.
P280	Wear eye protection / face protection.
P301+P312	IF SWALLOWED: Call a POISON CENTER/doctor if you feel unwell.
P330	Rinse mouth.
P304+P340	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337+P313	If eye irritation persists: Get medical advice/attention.
P403+P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.
P501	Dispose of contents/container in accordance with local/regional/national/international regulations.

(Contd.onpage 2)

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023
Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

(Contd. of page 1)

3 Composition and Information on Ingredients

Chemical Characterization:Mixtures
Description: Mixture of substances listed below with nonhazardous additions.

Hazardous Components:

CAS: 79-09-4	Propionic acid Flammable Liquids 3, H226; Acute Toxicity (Dermal) 3, H311; Skin Corrosion/Irritation 1B, H314; Acute Toxicity (Oral) 4, H302; STOT SE 3, H335	30-40%
CAS: 26027-38-3	Polyoxyethylated p-nonylphenol Acute Toxicity (Oral) 4, H302; Skin Corrosion/Irritation 2, H315; Eye Irritation 2A, H319	10-30%

4 First Aid Measures

Inhalation:If inhaled, remove to fresh air. Seek medical attention if breathing problems develop.

Skin Contact:
In case of skin contact, immediately remove contaminated clothing and wash affected areas with water and soap. Seek medical attention if irritation occurs.

Eye Contact:
In case of eye contact, rinse with water for several minutes, including under eyelids. Remove contact lenses, if present and easy to do. Continue rinsing for at least 15 minutes. Seek immediate medical attention.

Ingestion:
If swallowed, do not induce vomiting. Immediately rinse mouth with water. Never give anything by mouth to an unconscious person. Seek immediate medical attention.

Symptoms Caused by Exposure:
Inhalation: May cause respiratory irritation. May cause nasal irritation, sore throat, coughing, hoarseness and breathing difficulties.
Skin Contact: May cause skin irritation.
Eye Contact: Causes serious eye damage.
Ingestion: Harmful if swallowed. May cause gastrointestinal irritation, nausea, diarrhoea and vomiting.

5 Fire Fighting Measures

Suitable Extinguishing Media:Use fire extinguishing methods suitable to surrounding conditions.

Specific Hazards Arising from the Chemical:
Hazardous combustion products include oxides of carbon, oxides of nitrogen and oxides of phosphorous.
Product is not flammable.
Containers close to fire should be removed only if safe to do so. Use water spray to cool fire exposed containers.
Minimise run-off from fire fighting entering drains or water courses.

Special Protective Equipment and Precautions for Fire Fighters:

When fighting a major fire wear self-contained breathing apparatus and protective equipment.

6 Accidental Release Measures

Personal Precautions, ProtectiveEquipment and EmergencyProcedures:
Wear protective equipment. Keep unprotected persons away. Take care as spill may cause a slip hazard. Ensure adequate ventilation.

Environmental Precautions:

In the event of a major spill, prevent spillage from entering drains or water courses.

Methods and Materials for Containment and Cleaning Up:

Wipe up small spills with a paper towel or dry rag. Clean up all spills immediately, since spilled materials, even in small quantities, may present a slip hazard. Absorb large spills with sand, earth, vermiculite or some other absorbent material. Collect the spilled material and place into a suitable container for disposal. Decontaminate

(Contd. on page 3)

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023
Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

(Contd. of page 2)

spill area with water and detergents.

7 Handling and Storage

Precautionsfor Safe Handling:
Use of safe work practices are recommended to avoid eye or skin contact and inhalation of vapours or mists. Use only outdoors or in a well-ventilated area. Food, beverages and tobacco products should not be stored or consumed where this material is in use. Always wash hands before smoking, eating, drinking or using the toilet. Wash contaminated clothing and other protective equipment before storage or re-use. Provide eyewash fountains and safety showers in close proximity to points of potential exposure.

Conditions for Safe Storage:

Store in a cool, dry and well ventilated area. Keep container tightly closed when not in use. Protect from direct sunlight. Store at temperatures above 5 °C. Keep away from alkaline metals.

8 Exposure Controls and Personal Protection

Exposure Standards:
CAS: 79-09-4 Propionic acid
WESTWA: 30 mg/m³, 10 ppm

Engineering Controls:
Ensure adequate ventilation of the working area if the battery inner contents are exposed, keeping airborne concentrations below occupational exposure standards.

Respiratory Protection:

Natural ventilation should be adequate under normal use conditions.
Use an approved vapour respirator (Filter type A) under conditions where exposure to the substance is apparent (e.g. generation of high concentrations of mist or vapour, inadequate ventilation, development of respiratory tract irritation) and engineering controls are not feasible. See Australian Standards AS/NZS 1715 and 1716 for more information.

Skin Protection:

Chemical-resistant gloves. See Australian/New Zealand Standard AS/NZS 2161 for more information.
When selecting gloves for use against certain chemicals, the degradation resistance, permeation rate and permeation breakthrough time should be considered.
Occupational protective clothing (depending on conditions in which it has to be used, in particular as regards the period for which it is worn, which shall be determined on the basis of the seriousness of the risk, the frequency of exposure to the risk, the characteristics of the workstation of each worker and the performance of the protective clothing). See Australian/New Zealand Standard AS/NZS 4501 for more information.

Eye and Face Protection:

Safety glasses with top and side shields or goggles. See Australian/New Zealand Standards AS/NZS 1336 and 1337 for more information.

9 Physical and Chemical Properties

Appearance:		
Form:	Liquid	Dark brown
Colour:	Vinegar - lik e	No
Od ou r:	information available	
Odour Threshold:	approx. 3.6	No
pH-Value:	information available	No
Melting point/freezing point:	information available	
Initial Boiling Point/Boiling Range:	>120 °C	Not applicable
Flash Point:	No information available	
Flammability (solid, gas):	No information available	
Auto-ignition Temperature:		
Decomposition Temperature:		

(Contd. on page 4)

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023
Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

(Contd. of page 3)

Explosion Limits:	
Lower:	No information available No
Upper:	information available No
Vapour Pressure:	information available 1.02 -
Relative Density:	1.03 No information
Vapour Density:	available No information
Evaporation Rate:	available Forms an emulsion
Solubility in Water:	in water No information
Partition Coefficient (n-octanol/water):	available No information
Viscosity:	available

10Stability and Reactivity

Possibility of HazardousReactions: No dangerous reactions known under conditions of normal use.

Chemical Stability: Stable at ambient temperature and under normal conditions of storage and use.

Conditions to Avoid: Direct sunlight. Storage are temperatures below 5 °C.

Incompatible Materials:Alkaline materials.

Hazardous DecompositionProducts: Oxides of carbon, oxides of nitrogen and oxides of phosphorous.

11Toxicological Information

T oxicity:

LD50/LC50 Values:	
CAS: 79-09-4 Propionic acid	
Oral	LD502,600 mg/kg (Rattus norvegicus (rat))
	LD50500 mg/kg (Oryctolagus cuniculus (rabbit))
CAS: 26027-38-3Polyoxyethylated p-nonylphenol	
Oral	LD504,290 mg/kg (Mus musculus (mouse))
	1,410 mg/kg (Rattus norvegicus (rat))
	LD502,830 mg/kg (Oryctolagus cuniculus (rabbit))

Acute Health Effects
Inhalation:
May cause respiratory irritation. May cause nasal irritation, sore throat, coughing, hoarseness and breathing dif f ic ulties.
Skin:May cause skin irritation.
Eye:Causes serious eye irritation.
Ingestion:Harmful if swallowed. May cause gastrointestinal irritation, nausea, diarrhoea and vomiting.

Skin Corrosion / Irritation:Based on classification principles, the classification criteria are not met.

Serious Eye Damage / Irritation:Causes serious eye irritation.

Respiratory or Skin Sensitisation:Based on classification principles, the classification criteria are not met.

Germ Cell Mutagenicity:Based on classification principles, the classification criteria are not met.

Carcinogenicity:Based on classification principles, the classification criteria are not met.

Reproductive Toxicity:Based on classification principles, the classification criteria are not met.

Specific Target Organ Toxicity (STOT) - Single Exposure:May cause respiratory irritation.

Specific Target Organ Toxicity (STOT) - Repeated Exposure:
Based on classification principles, the classification criteria are not met.

Aspiration Hazard:Based on classification principles, the classification criteria are not met.

(Contd. on page 5)

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023
Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

(Contd. of page 4)

Chronic Health Effects: No information available

Existing Conditions AggravatedbyExposure: No information available

12Ecological Information

Ecotoxicity:

Aquatic toxicity:
The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.

Persistence and Degradability:No data available on finished product.

Bioaccumulative Potential:No data available on finished product.

Mobility in Soil:No data available on finished product.

Other adverse effects:No further relevant information available.

13Disposal Considerations

Disposal Methods and Containers:Dispose according to applicable local and state government regulations.

Special Precautions for Landfill or Incineration:
Please consult your state Land Waste Management Authority for more information.

14Transport Information

UN Number	Not regulated
Proper Shipping Name	Not regulated
Dangerous Goods Class	Not regulated
Packing Group:	Not regulated

15Regulatory Information

AustralianInventoryofIndustrial Chemicals:
All ingredients are listed.

Standard fortheUniform Scheduling ofMedicinesandPoisons(SUSMP) - PoisonSchedule:
Poisons Schedule: 5

Australian Pesticidesand Veterinary MedicinesAuthority:
This product is registered with the Australian Pesticides and Veterinary Medicines Authority. APVMA approval number 66985.

16Other Information

Date of Preparation or Last Revision: 06.09.2023

Prepared by:MSDS.COM.AU Pty Ltd www.m sds .c om .au

Abbreviations and acronyms:
GHS: Globally Harmonised SystemofClassification and Labelling of Chemicals
CAS: Chemical Abstracts Service (division of the American Chemical Society)
LC50: Lethal concentration, 50 percent
LD50: Lethal dose, 50 percent
IARC: International Agency for Research on Cancer
STEL: Short Term Exposure Limit
TWA: Time Weighted Average
NES: National Exposure Standard (Safe Work Australia - Workplace Exposure Standards For Airborne Contaminants)
Flammable Liquids 3: Flammable liquids – Category 3
Acute Toxicity (Oral) 4: Acute toxicity – Category 4
Acute Toxicity (Dermal) 3: Acute toxicity – Category 3
Skin Corrosion/Irritation 1B: Skin corrosion/irritation – Category 1B

(Contd. on page 6)

Safety Data Sheet
according to WHSRegulations

Print date: 22.09.2023 Revision date: 06.09.2023
Product Name:LOVELAND PRODUCTS LI-700 SURFACTANT

(Contd. of page 5)

Skin Corrosion/Irritation 2: Skin corrosion/irritation – Category 2
Eye Irritation 2A: Serious eye damage/eye irritation – Category 2A
STOT SE3: Specific target organ toxicity (single exposure) – Category 3

Disclaimer

This SDS is prepared in accord with the Safe Work Australia document “Code of Practice for the Preparation of Safety Data Sheets for Hazardous Chemicals - July 2020”.
The information contained in this safety data sheet is provided in good faith and is believed to be accurate at the date of issuance. Nutrien Ag Solutions makes no representation of the accuracy or comprehensiveness of the information and to the full extent allowed by law excludes all liability for any loss or damage related to the supply or use of the information in this material safety data sheet. MSDS.COM.AU Pty Ltd is not in a position to warrant the accuracy of the data herein. The user is cautioned to make their own determinations as to the suitability of the information provided to the particular circumstances in which the product is used.